FOR BHADRI · BACKGROUND ONLY; YOU DO NOT NEED IT TO RUN THE TEST KIT

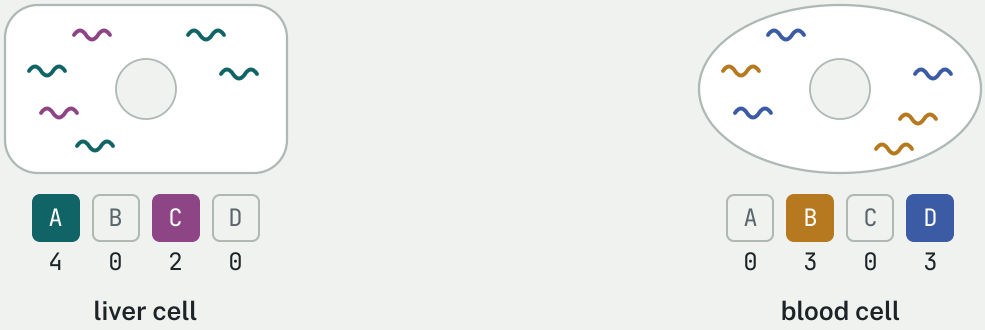
AI bioinformatician: what we are trying to do

Turning biological data into conclusions needs a bioinformatician, who can both write the analysis code and judge what the results mean. They are scarce, so experiments wait. We want to know whether an AI agent can do that job well enough for a lab to trust it and plan its next experiments.

Biology in five minutes

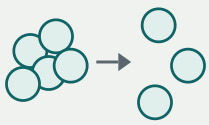
Cells, genes and RNA

Every cell in a fish has the same 26,000 or so genes, like the same codebase with different parts active. An active gene is copied into RNA messages, so counting a cell's messages tells you what the cell is and what it is doing.



Same four genes, different ones active. Each squiggle is one RNA message; the numbers are the counts.

Single-cell RNA sequencing



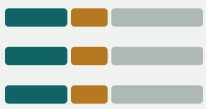
Separate

Tissue is broken into single cells.



One cell, one droplet

Each cell meets a bead with its own barcode.



Sequence

Each sequenced piece, called a read, carries the cell's barcode, a UMI and a piece of a gene.

barcode UMI gene

gene	cell 1	cell 2
A	4	0
B	0	3
C	2	0
D	0	3

Count

One row per gene, one column per cell.

The real table has a row for each of tens of thousands of genes and thousands of columns, mostly zeros: many genes are off and most messages are missed. Copies made for sequencing share a UMI and count once, like de-duplicating retried requests by request ID.

What the analyst does with the table



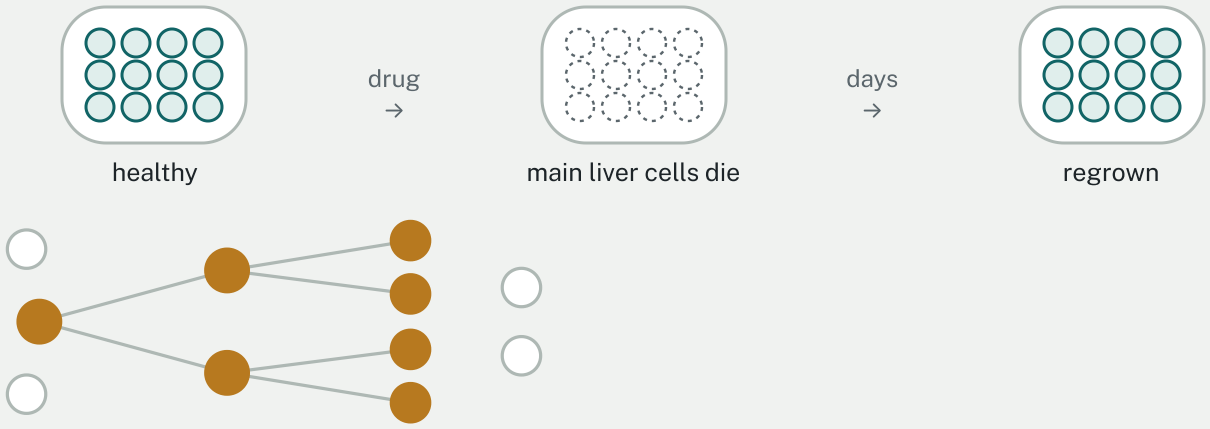
1. Remove what is not a real cell (crosses): empty droplets, droplets that caught two cells, dying cells.
2. Group similar cells into clusters (colours).
3. Name each cluster by its marker genes, the genes switched on mainly in that cluster. This needs biology.
4. Compare groups and time points: genes and pathways (sets of genes that work together) going up or down, cells ordered along a process, which cells signal to which.
5. Say what it means, how sure it is, and what experiment would test it.

Each dot is a cell. Cells whose gene counts follow a similar pattern sit together. Colours and numbers here mark clusters, not genes.

Running the tools is routine. The hard part is judgment: technical noise, such as samples processed on different days, can pass for a discovery. That is what we test.

The animal: zebrafish liver regeneration

The zebrafish liver can regrow after serious damage. In the fish in this study, the main liver cells carry an engineered enzyme that turns a drug into a poison, so the drug kills only those cells. The liver is then sampled on set days as it recovers.



Top: injury and recovery. Bottom: lineage tracing. A chosen set of cells carries a permanent colour that every descendant inherits, so you can tell what they became.

The test

We test the agent where the answers are known: a published study from this field, used like a test set with a reviewed answer key.

ANALYST AGENT: NO PAPER, NO INTERNET

1. Gets the data and the experiment design.
2. Part 1: open questions. Answers are frozen.
3. Part 2: targeted questions.
4. Part 3: proposes next experiments.

GRADING: THE ANALYST NEVER SEES THIS SIDE

5. Two grader agents score it against the answer key, built mostly from findings the authors confirmed with lab experiments.
6. A reconciler merges the two gradings.
7. Prasanna decides the open points.
8. Output: a scorecard, and the agent's figures beside the paper's.

Pass bar: at least 3 of 6 core findings fully recovered, with no major false claims. Run A starts from the count table. Run B starts from the raw reads, so the agent also has to build the table.

One limit

The paper is public, so a model may have read it in training. Blinding stops look-ups, not memory. Requiring evidence for every claim makes remembered answers easier to spot, but the clean test comes later, on unpublished data, after Prasanna checks with the institution.

Why it matters

If it works, labs can plan experiments sooner. In the closest public test (scBench-Long, LatchBio, 2026) the best agent passed about 1 run in 4. We found no test on zebrafish, on regeneration, or scored against lab-confirmed findings, though our search was not exhaustive. Details: `background/landscape.md`.

Words you will see in the kit

Hepatocyte
The main working cell of the liver.

dpa
Days post ablation: days since the drug treatment that killed the main liver cells.

Drop-seq
The droplet-and-bead method used to make this dataset.

UMI
Unique molecular identifier: a short random tag on each captured RNA message, so copies made for sequencing count once.

Biliary epithelial cell (BEC)
A cell lining the bile ducts, the liver's drainage tubes.

FASTQ
The raw text files of sequencer reads.

GEO
The public NCBI archive where the data are stored.

Background only. Never give this page to the analyst agent. Questions about the biology go to Prasanna.